

Redox signalling in apoptosis and cellular differentiation : the ASK-p38 pathway

UE MMSC
Master BI

Isabelle Petropoulos
isabelle.petropoulos@sorbonne-universite.fr

Aurore L'Honoré
aurore.lhonore@sorbonne-universite.fr

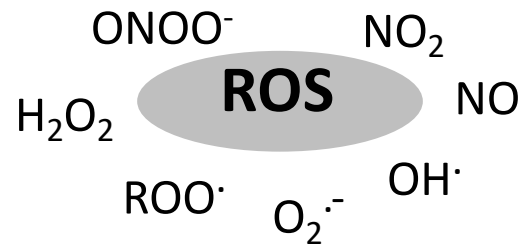
Reactive Oxygen Species: Production

ENDOGENOUS SOURCES

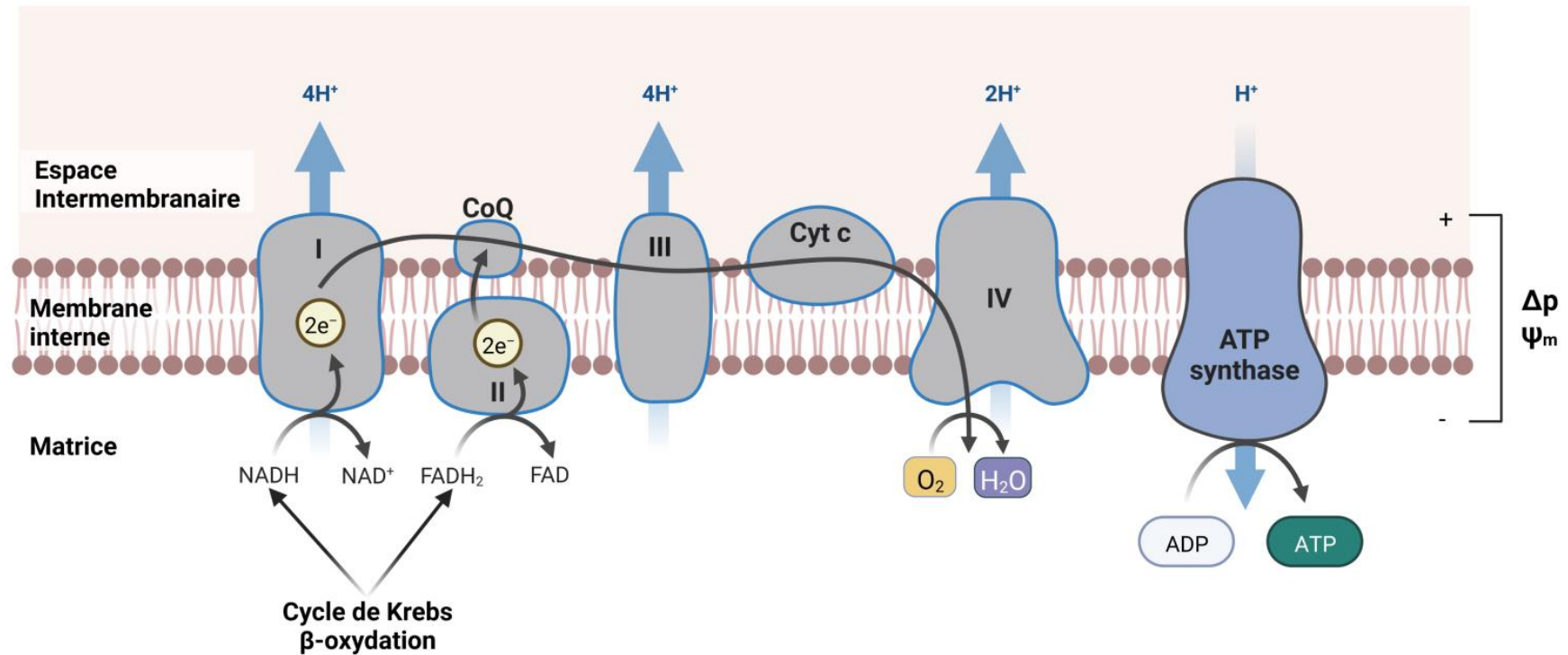
Mitochondria
Peroxisomes
Lipoxygenases
NADPH oxydases
Cytochrome P450

EXOGENOUS SOURCES

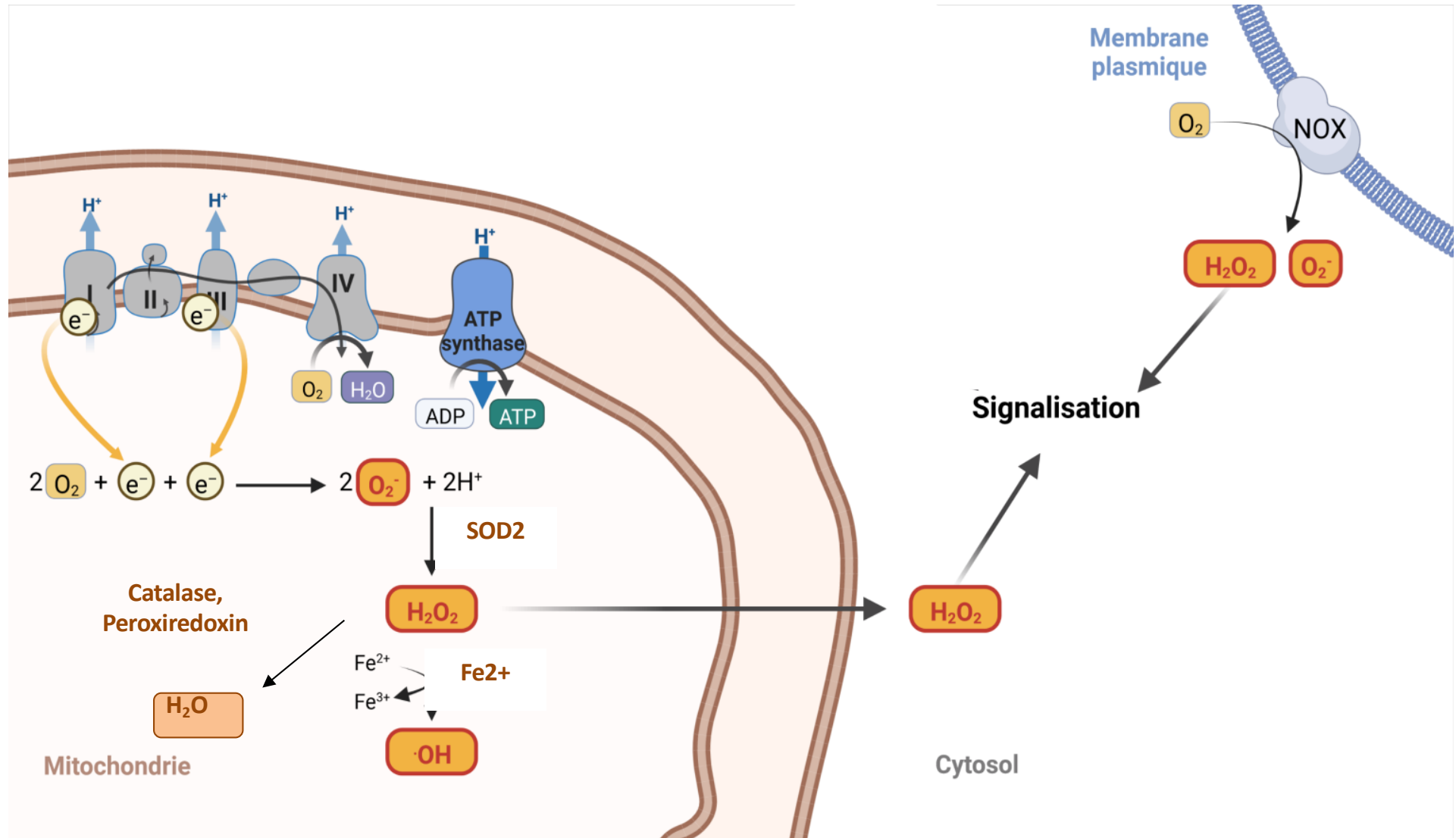
UV, ionizing radiations,
chemotherapeutics,
Inflammation



Reactive oxygen species : Production



Reactive oxygen species : Production



Reactive oxygen species : Production and scavenging

ENDOGENOUS SOURCES

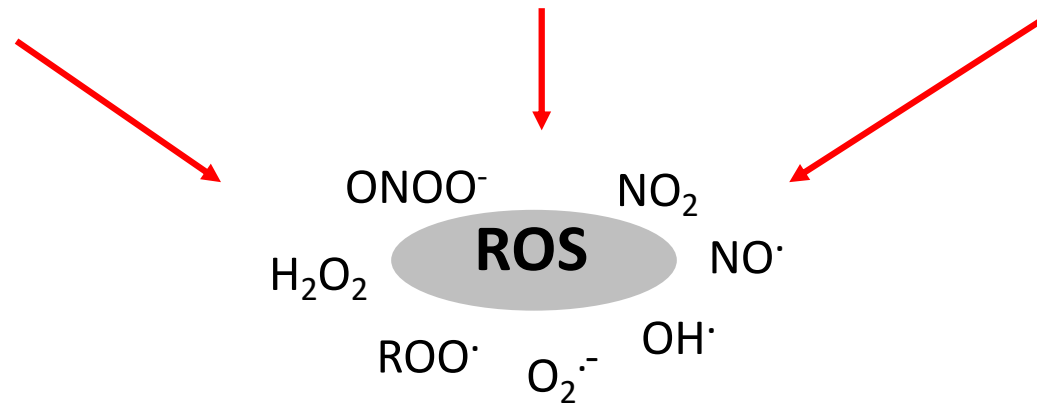
Mitochondria
Peroxisomes
Lipoxygenases
NADPH oxydases
Cytochrome P450

ANTIOXIDANT DEFENCES

Enzymatic systems :
CAT, SOD, GPx, Prx, Trx
Non-Enzymatic systems :
Glutathion, vitamines (A, C et E)

EXOGENOUS SOURCES

UV, ionizing radiations,
chemotherapeutics,
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Reactive oxygen species : Production and scavenging

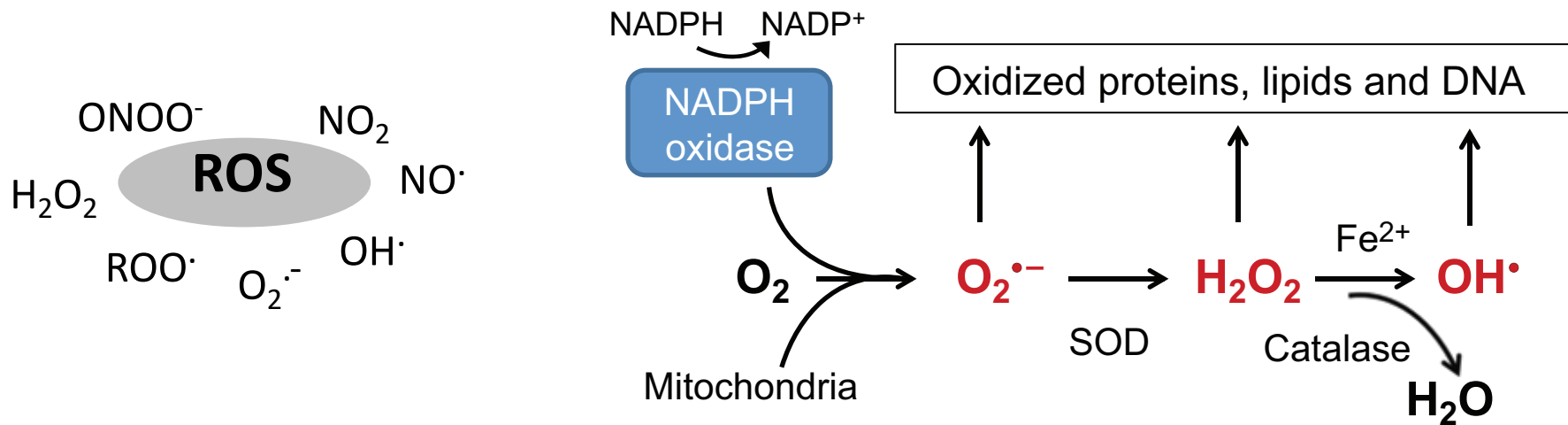
ENDOGENOUS SOURCES

Mitochondria
NADPH oxydases

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EXOGENOUS SOURCES



Reactive oxygen species : Production and scavenging

ENDOGENOUS SOURCES

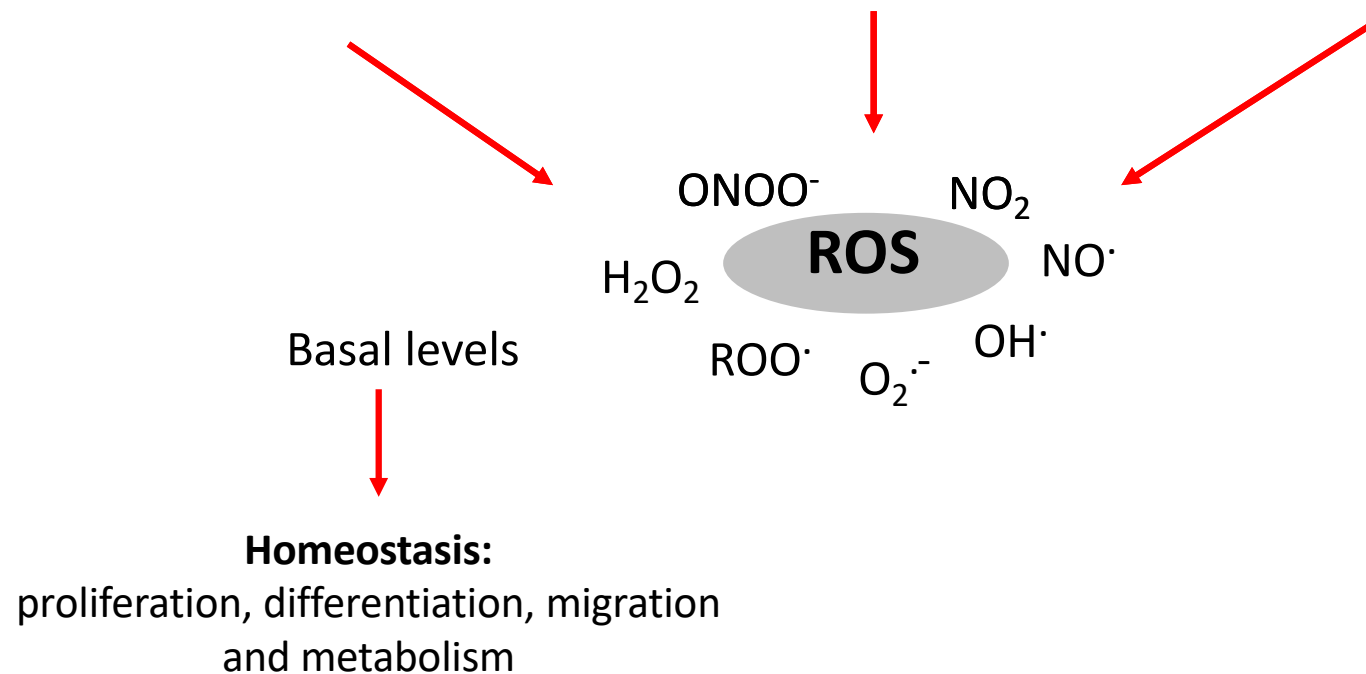
Mitochondria
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EXOGENOUS SOURCES

UV, ionizing radiations,
chemotherapeutics,
Inflammation



Espèces Réactives de l'Oxygène: Production et Dégradation

SOURCES ENDOGÈNES

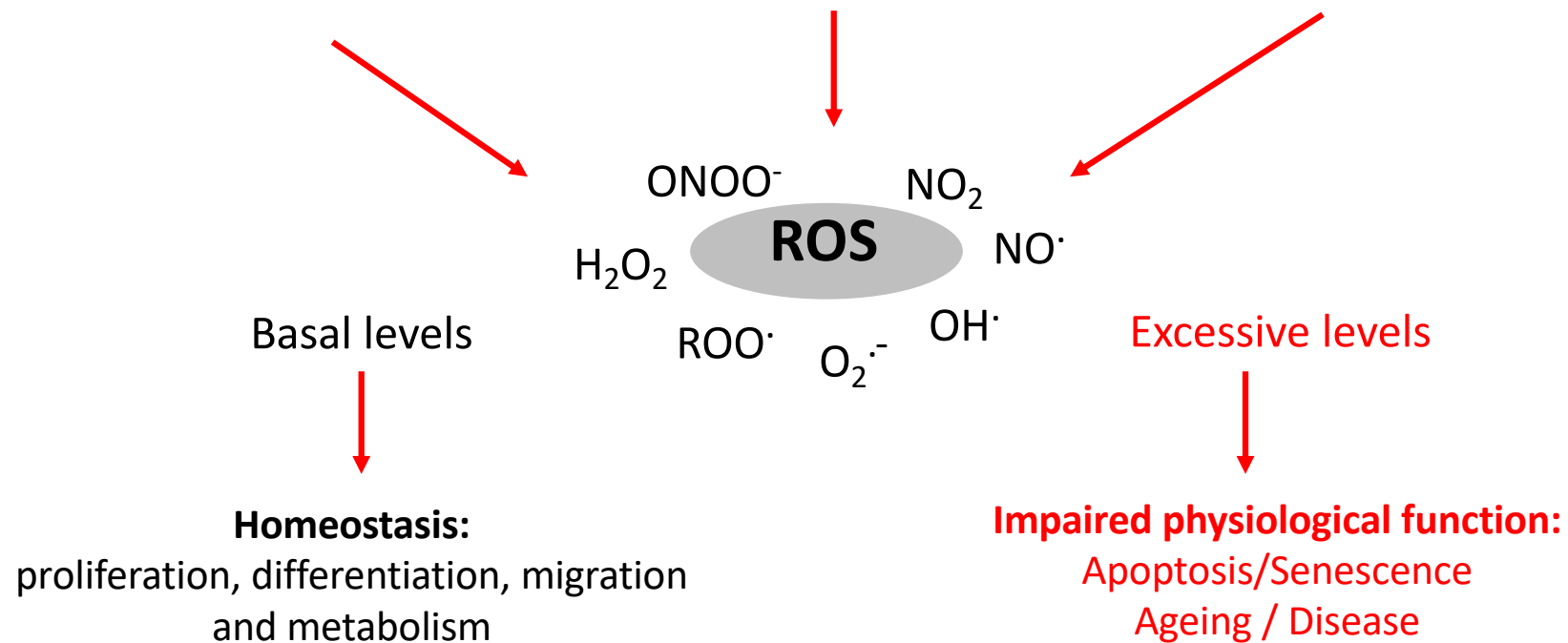
Mitochondries
Péroxyosomes
Lipoxygénases
NADPH oxydases
Cytochrome P450

DÉFENSES ANTIOXYDANTES

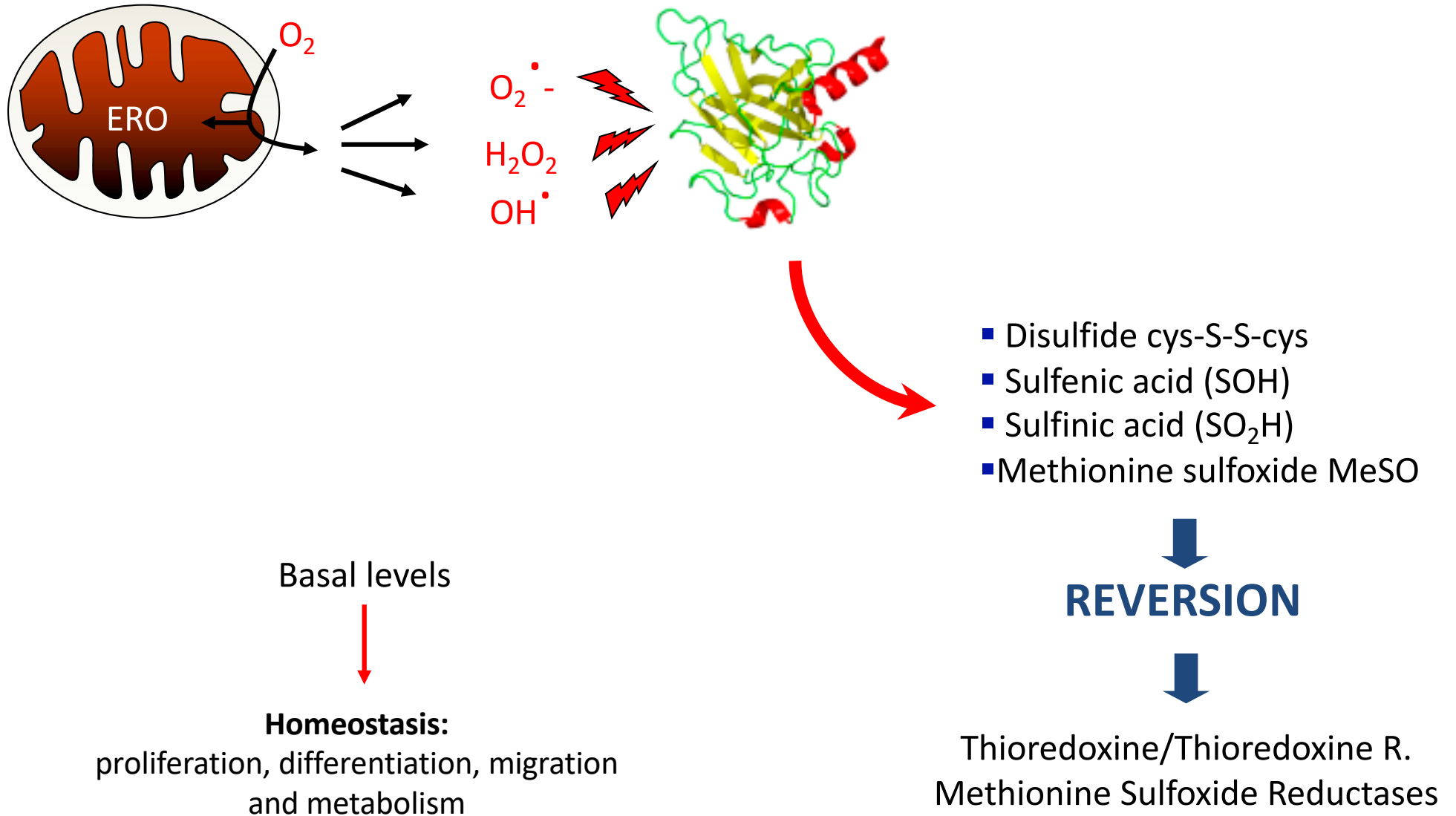
Systèmes enzymatiques :
CAT, SOD, GPx, Prx, Trx
Systèmes non-enzymatiques :
Glutathion, vitamines (A, C et E)

SOURCES EXOGÈNES

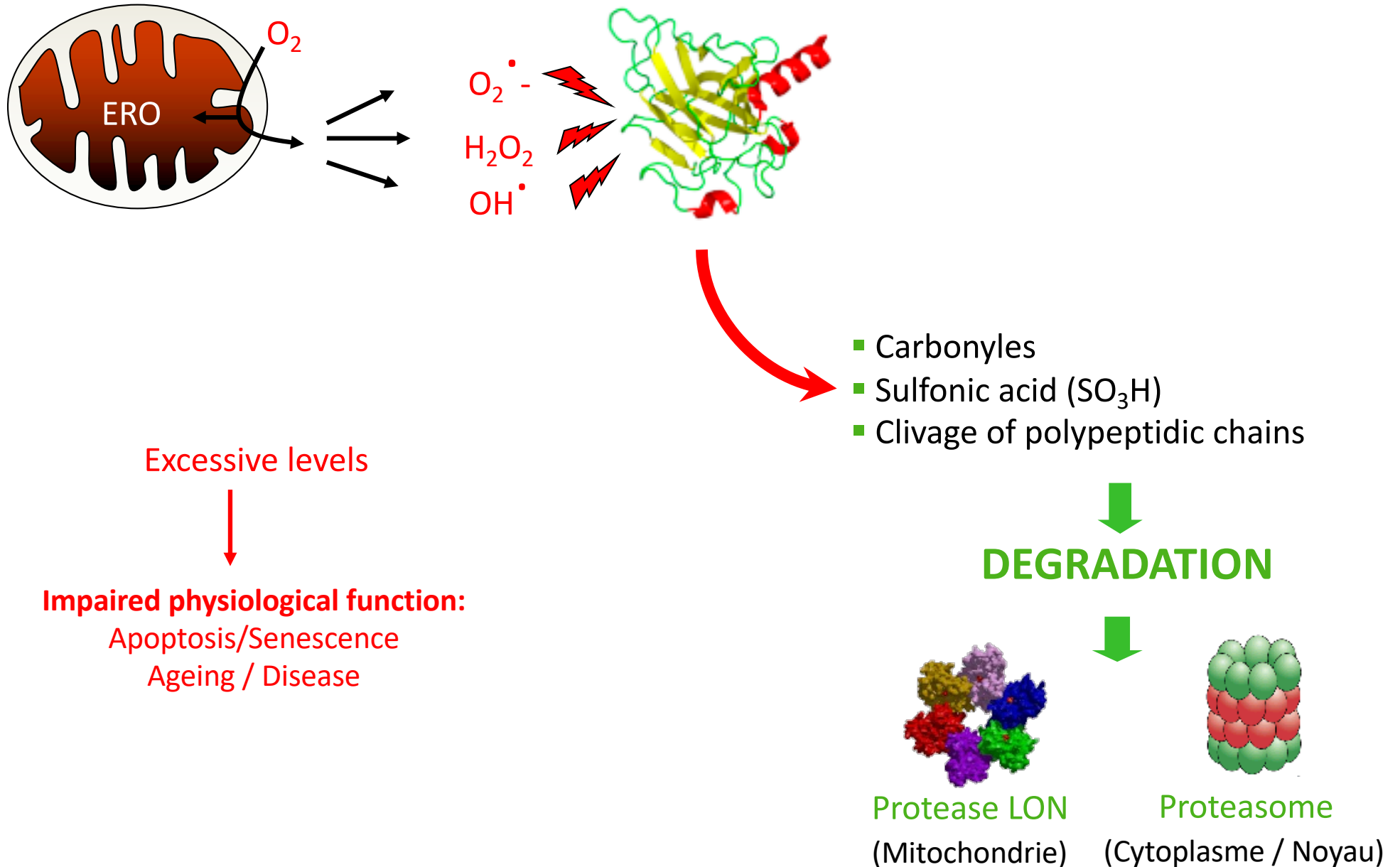
UV, Radiations ionisantes
Polluants
Xénobiotiques oxydants
Inflammation



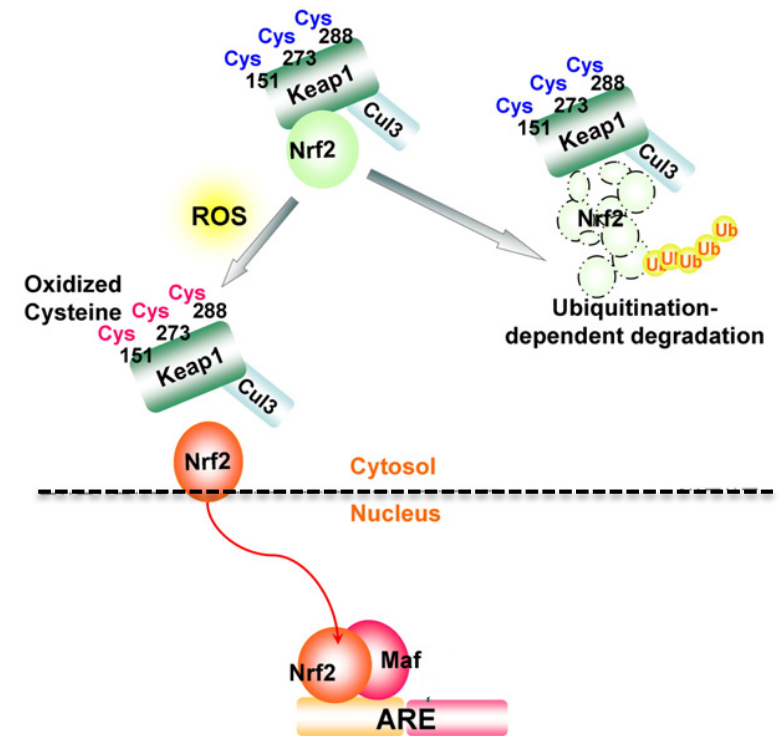
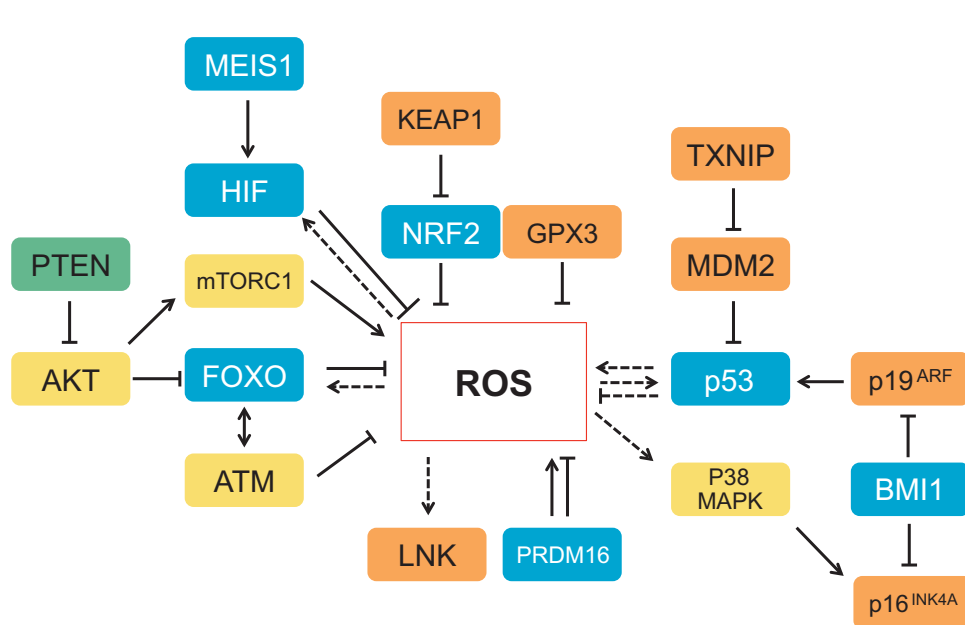
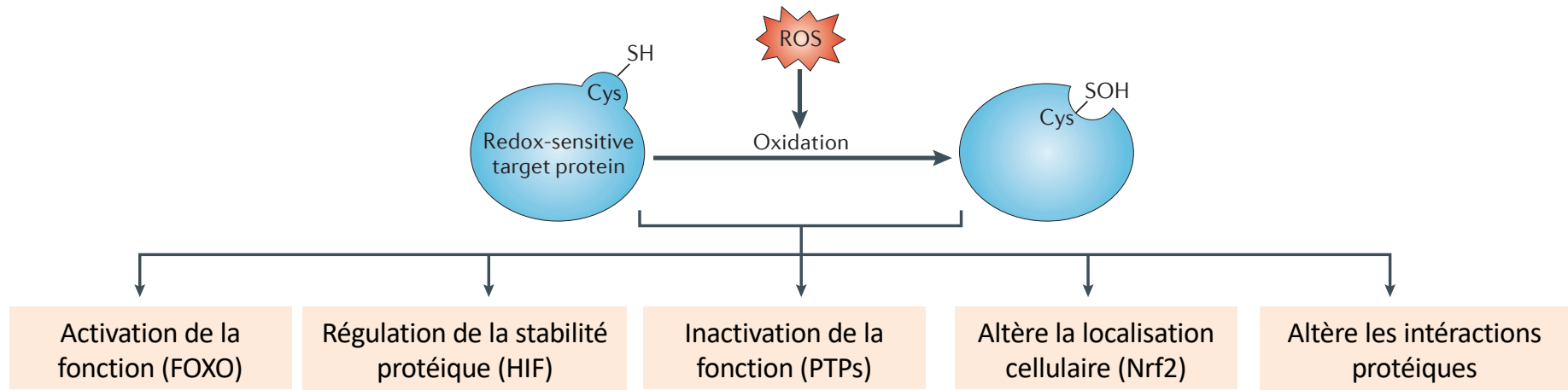
Protein oxidation and reduction



Oxydation des protéines et systèmes de maintenance



Redox-mediated signalling pathways :



Redox-dependent transcription factors

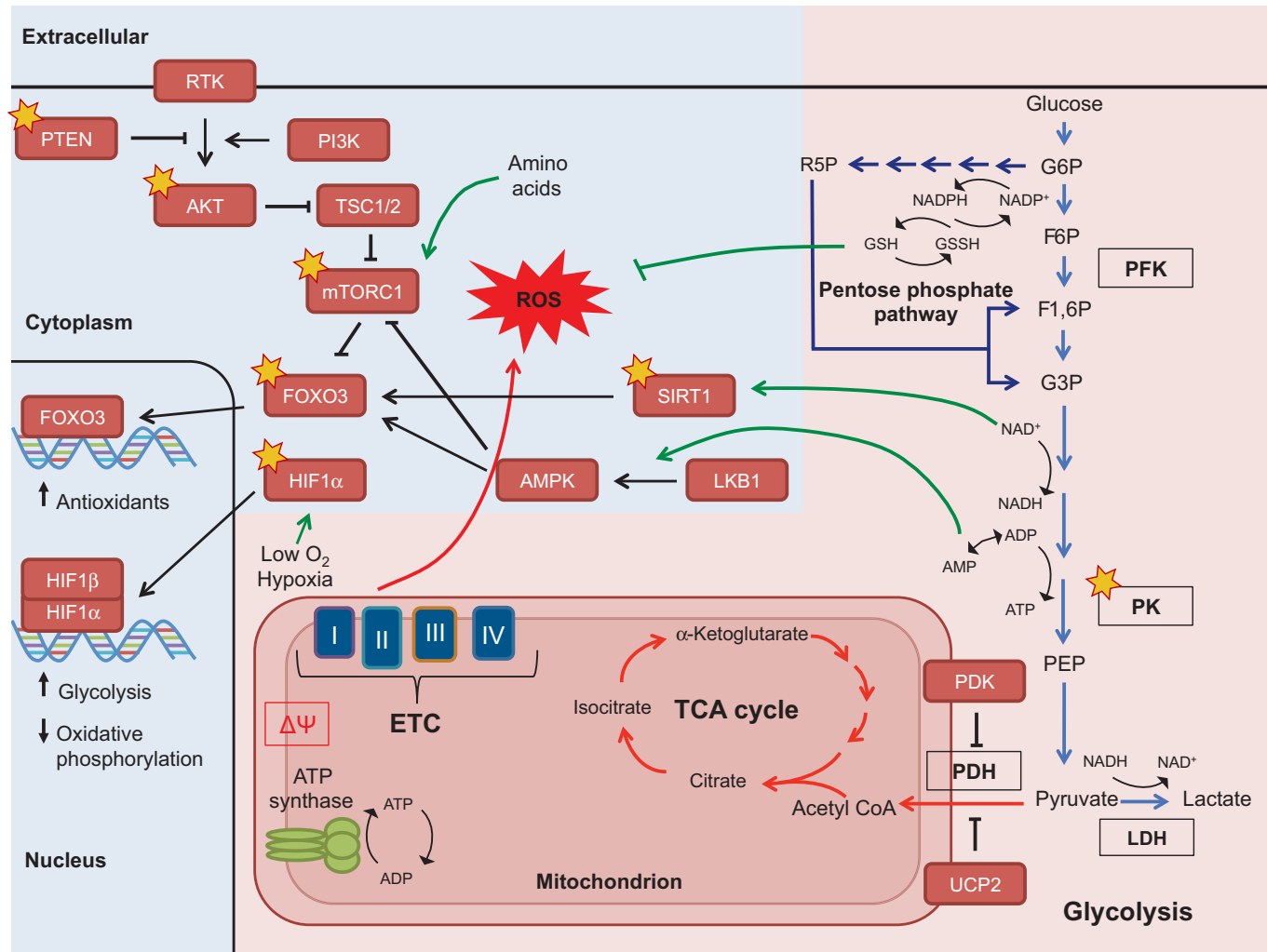
Redox-dependent kinases

Redox-dependent phosphatases

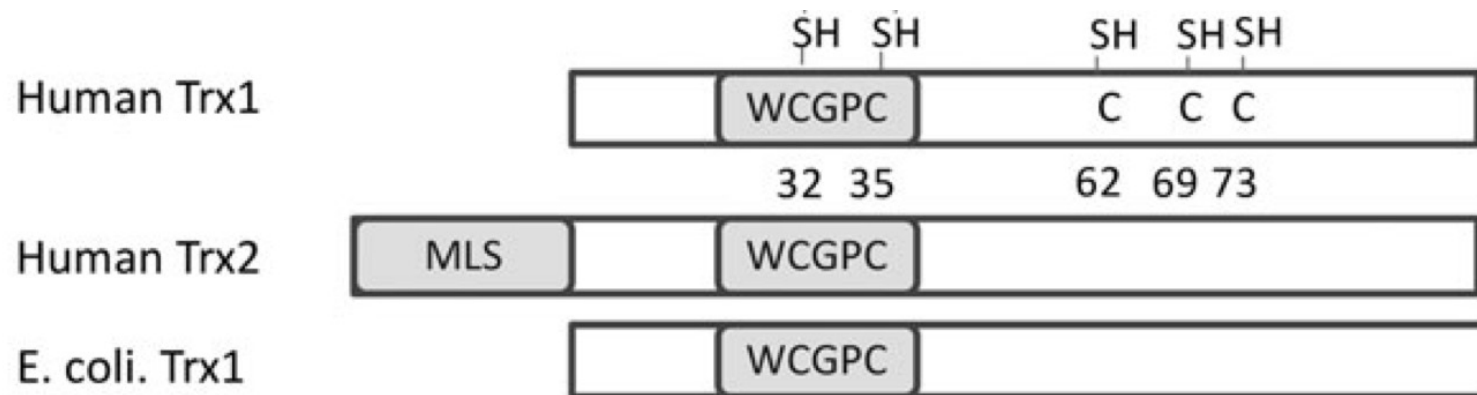
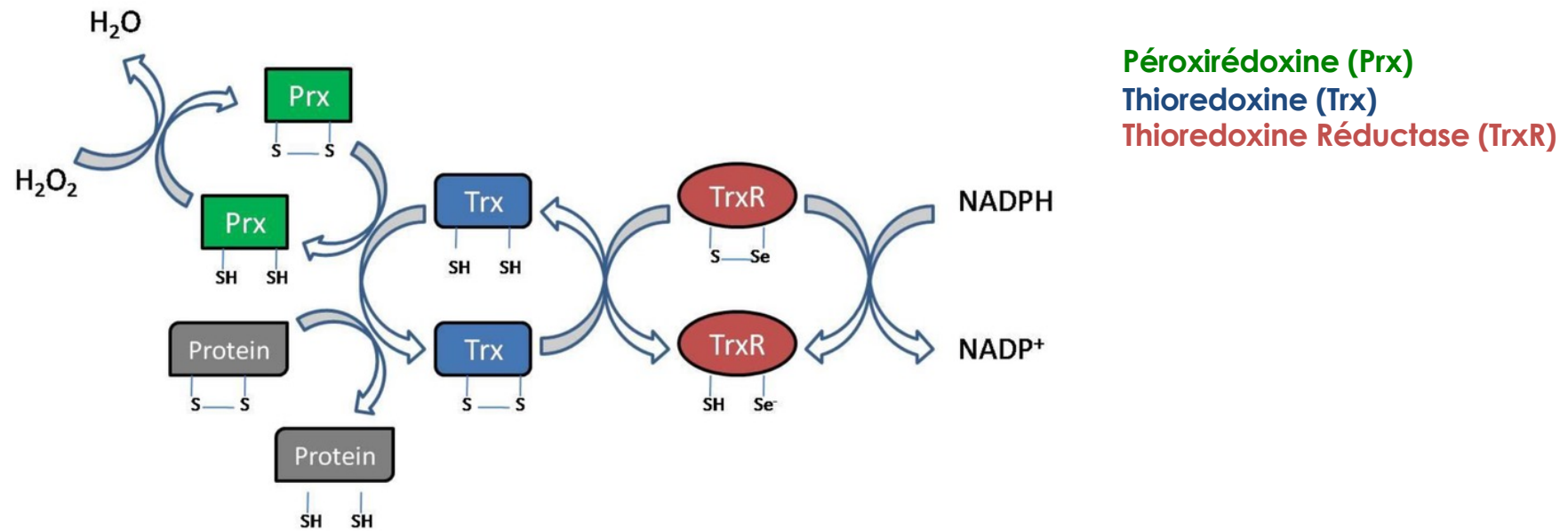
Regulators of redox state

Antioxidant Response Element (ARE)

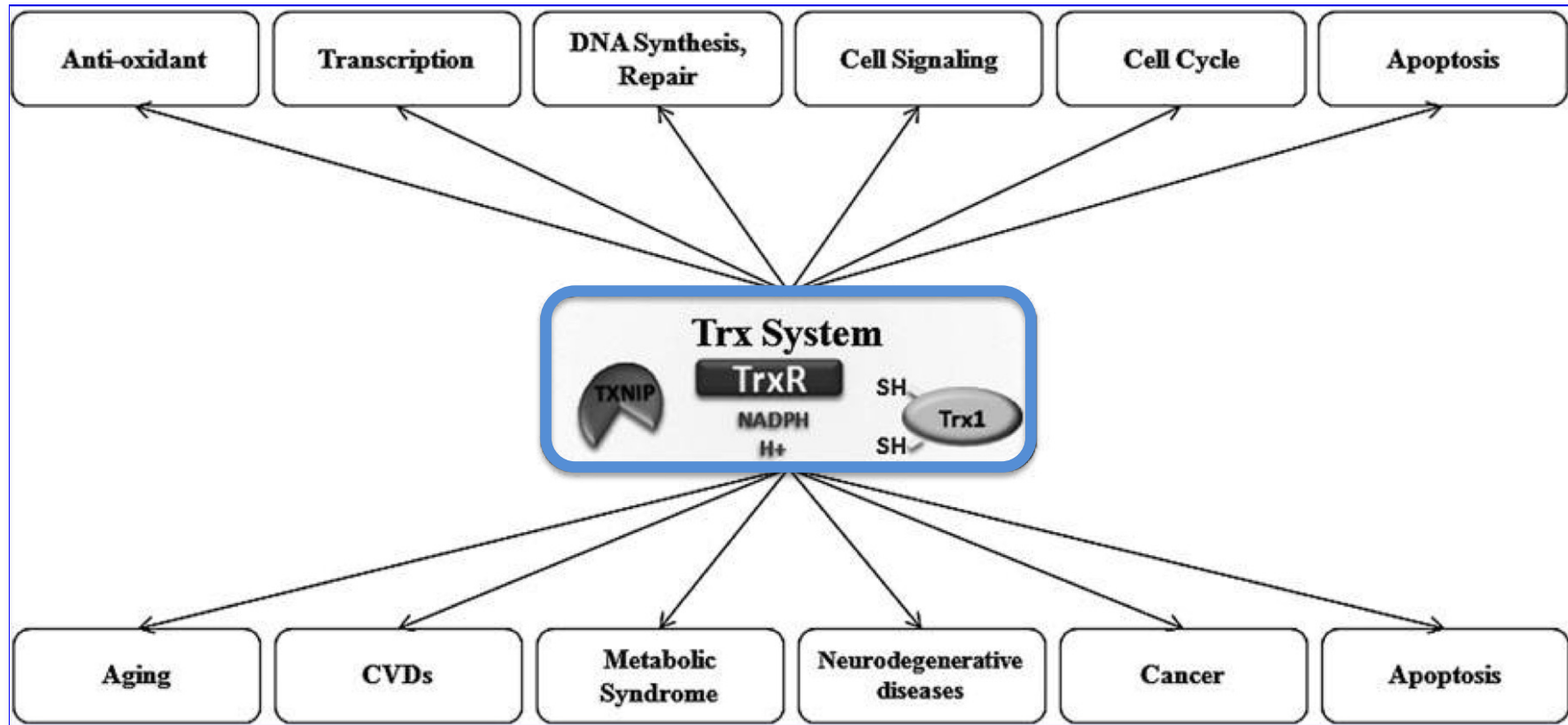
Redox-mediated signalling pathways :



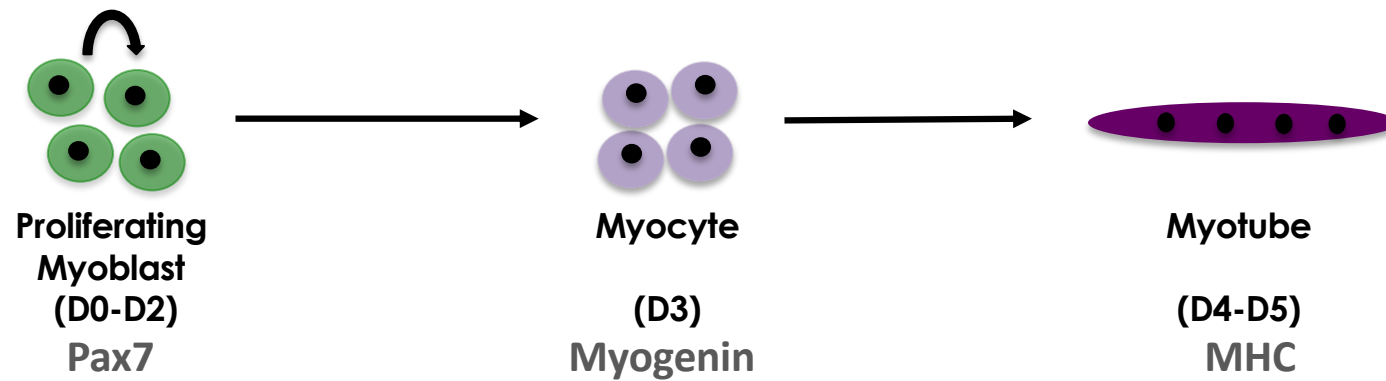
Thioredoxin roles and structure



Thioredoxin roles and structure



In vitro model of myogenic differentiation



Pax7, Myogenin: Transcription Factors
MHC: Cytoskeleton protein

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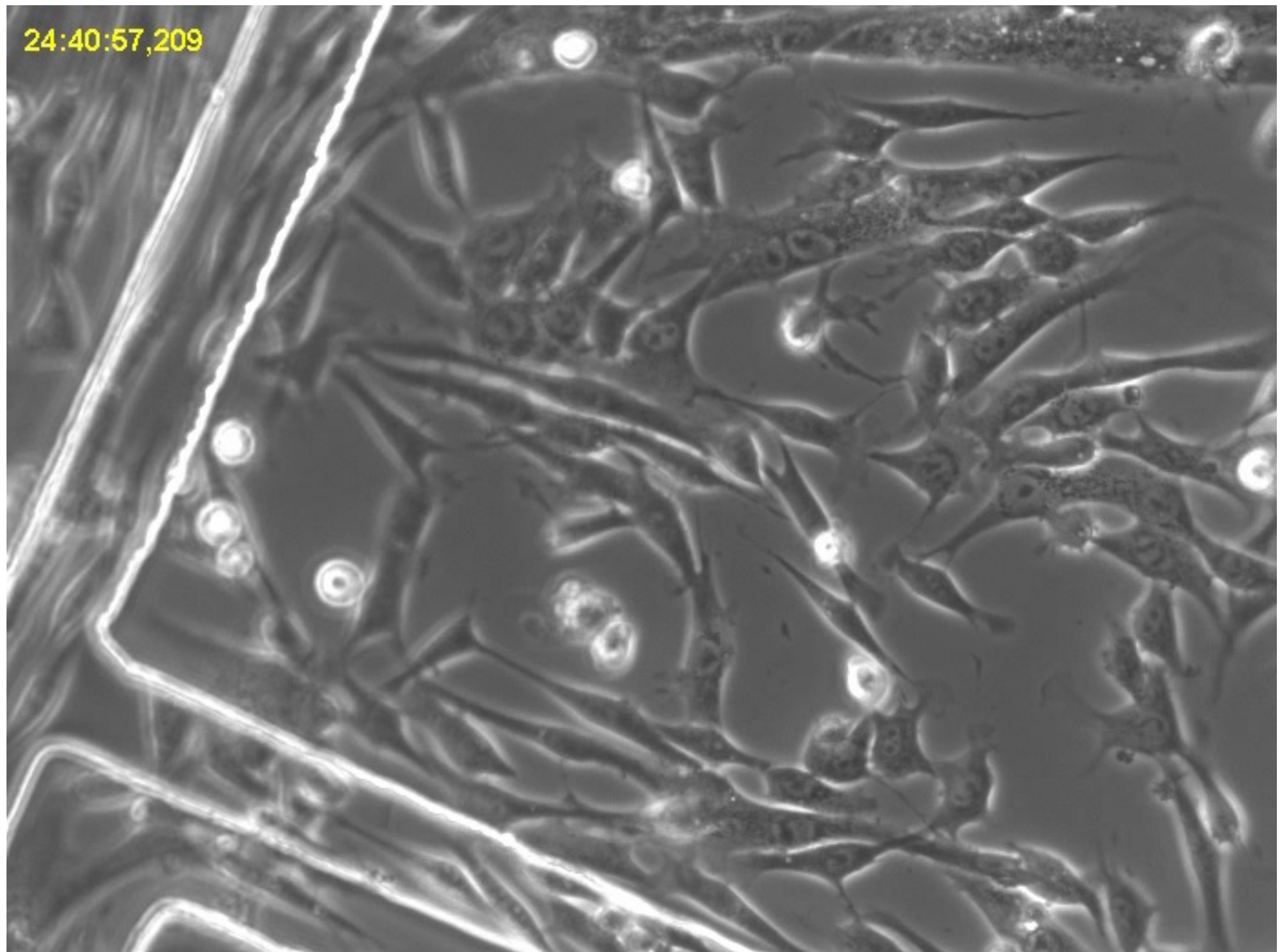
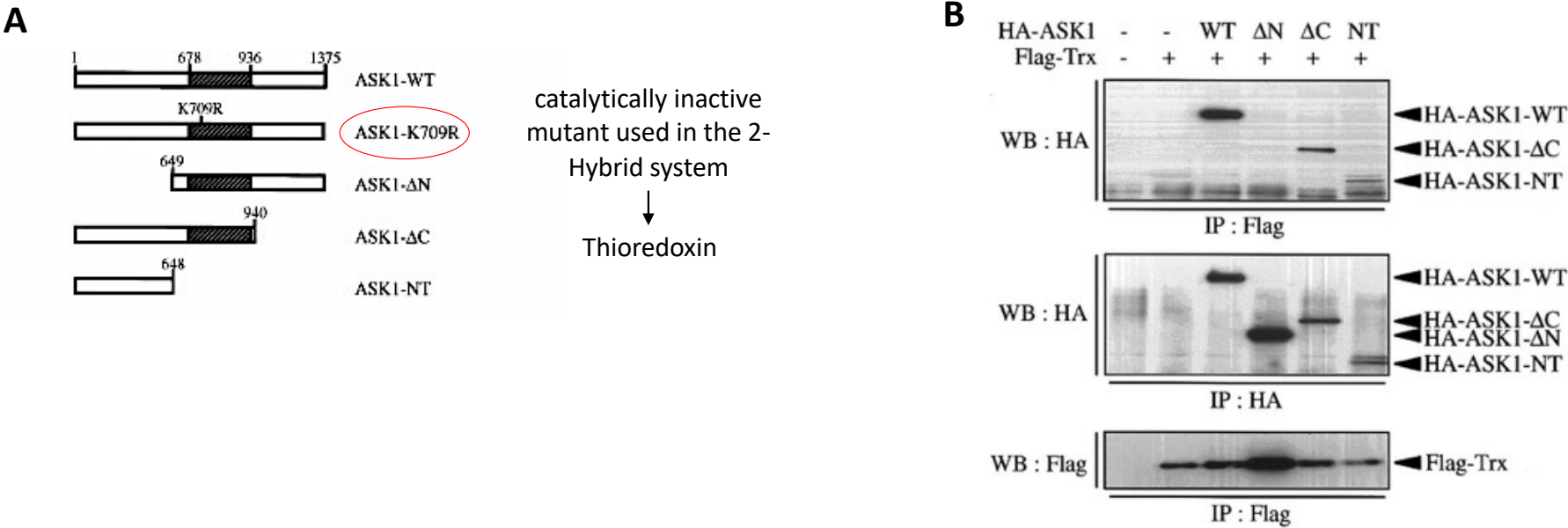


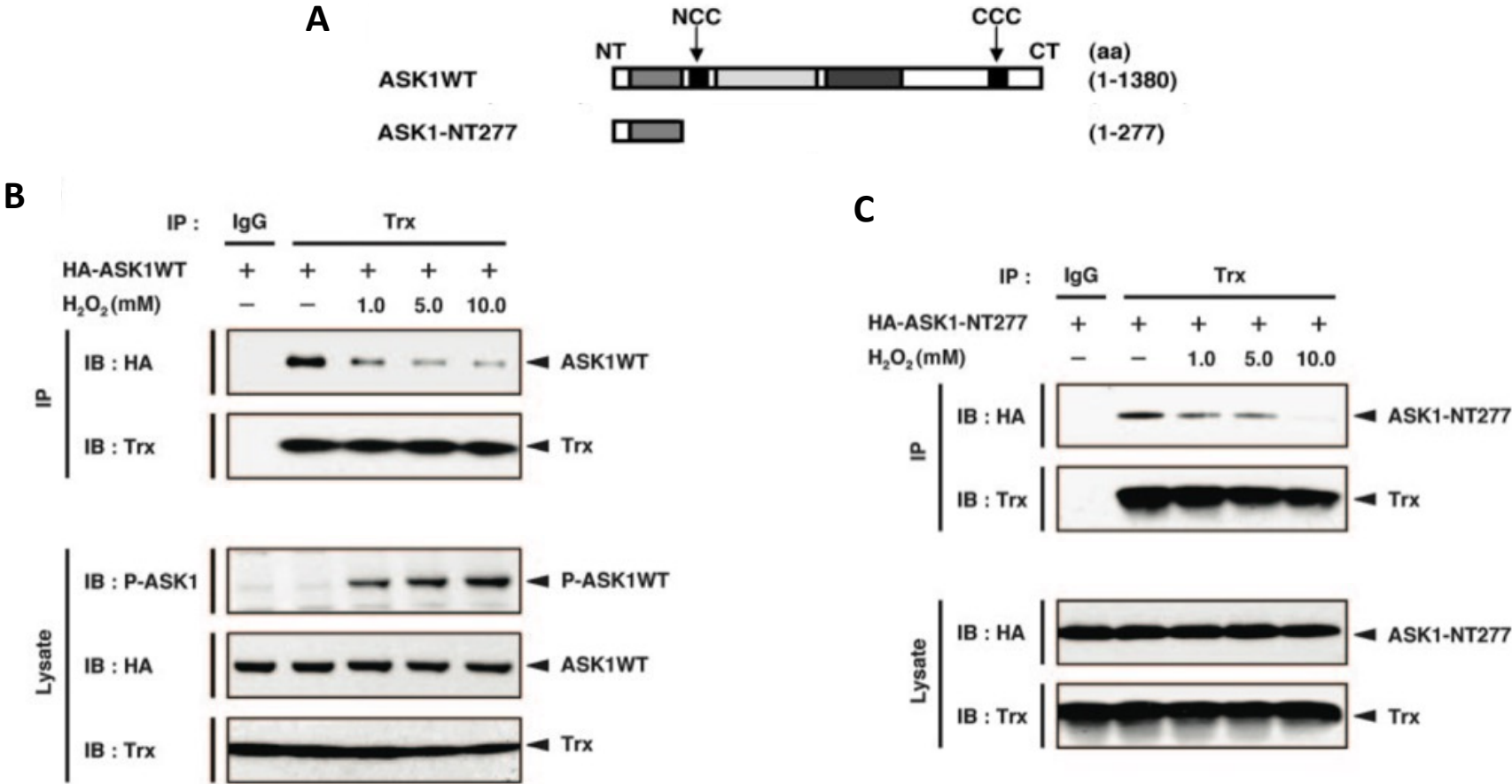
Figure 1a



(A) Schematic representation of mouse ASK1WT and its deletion mutants, with amino acid numbers indicated. The NCC and CCC domains are indicated. Interaction of ASK1-K709R with Trx in yeast. **(B)** HEK293A cells were cotransfected with Flag-Trx and either ASK1WT, ASK1ΔC, ASK1ΔN and ASK1-NT. Cell lysates were immunoprecipitated with anti-Flag antibody. Immunoprecipitates and aliquots of each lysate were subjected to SDS-PAGE followed by immunoblotting with anti-HA and anti-Flag antibodies. IP, immunoprecipitation; IB, immunoblotting.

(Saitoh et al., 1998)

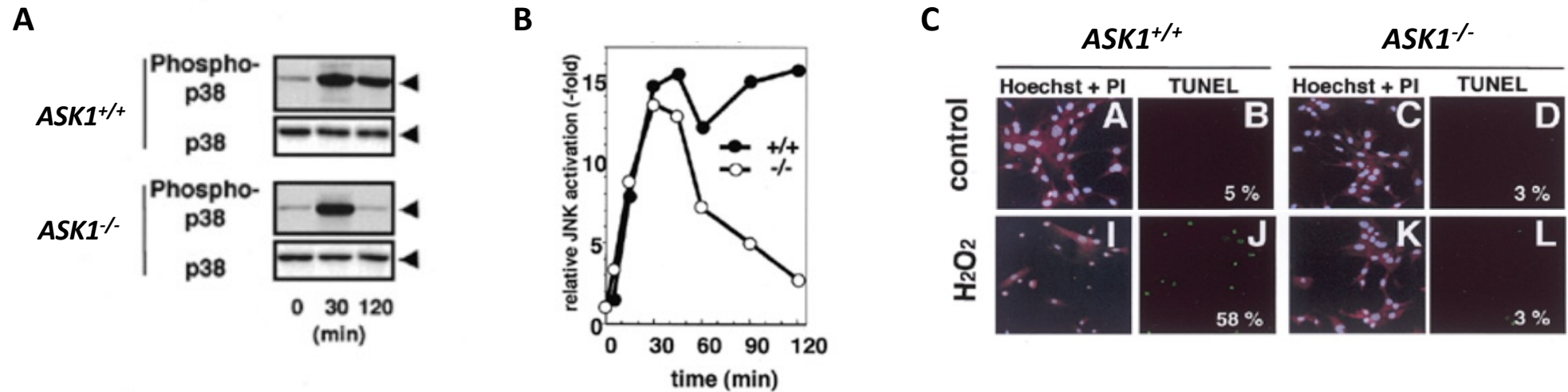
Figure 1b



(A) Schematic representation of mouse ASK1WT and its C-terminal deletion mutant, with amino acid numbers indicated. **(B, C)** HEK293A cells transfected with the indicated combinations of HA-tagged ASK1-WT (B) and ASK1-NT277 (C) were treated with indicated concentrations of H₂O₂. Cell lysates were immunoprecipitated with anti-Trx or anti-IgG antibody. Immunoprecipitates and aliquots of each lysate were subjected to SDS-PAGE followed by immunoblotting with anti-HA, anti-Trx and anti-phospho-ASK1 antibodies. IP, immunoprecipitation; IB, immunoblotting.

(Fujino et al., 2007)

Figure 1c



(A) Mouse Embryonic Fibroblasts (MEFs) purified from control and ASK1 knock-out embryos were treated with 0.1 mM of H₂O₂ for indicated time. Cell lysates were subjected to western blot analysis with antibodies specifically recognizing the activated forms of p38 (phospho-p38). Membranes were reprobed with antibodies to total p38 as loading controls. **(B)** Graphic representation of **(A)**. Fold activation was calculated by dividing the intensity of phosphorylated p38 at different time points by that at time zero (representative of three independent experiments). **(C)** Cell morphology (**A**, **C**, **I**, **K**) was analyzed by fluorescence double-staining with Hoechst 33258 (blue) and PI (red) of control and ASK1 knock-out MEFs that were untreated (**A** and **C**), treated with 0.2 mM H₂O₂ plus 25 mM aminotriazole (catalase inhibitor) for 2h (**I** and **K**). TUNEL staining (green; **B**, **D**, **J**, **L**) of the same fields is shown with numbers which represent percentages of TUNEL-positive cells per total cell count. At least 100 cells from three random fields were counted in each experiment, and data are from one of more than three representative experiments.

Figure 1d

Redox regulation of ASK1 activity

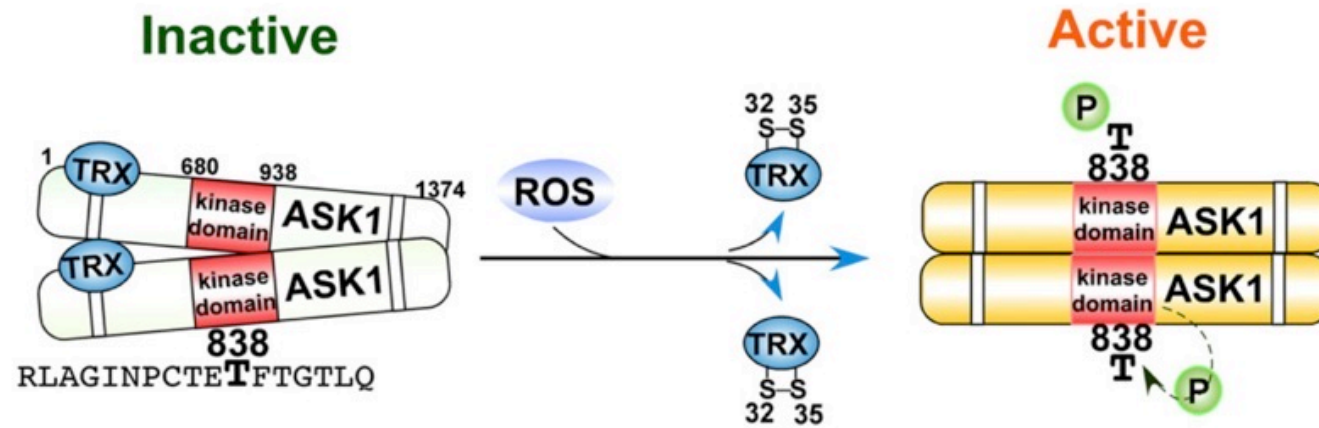
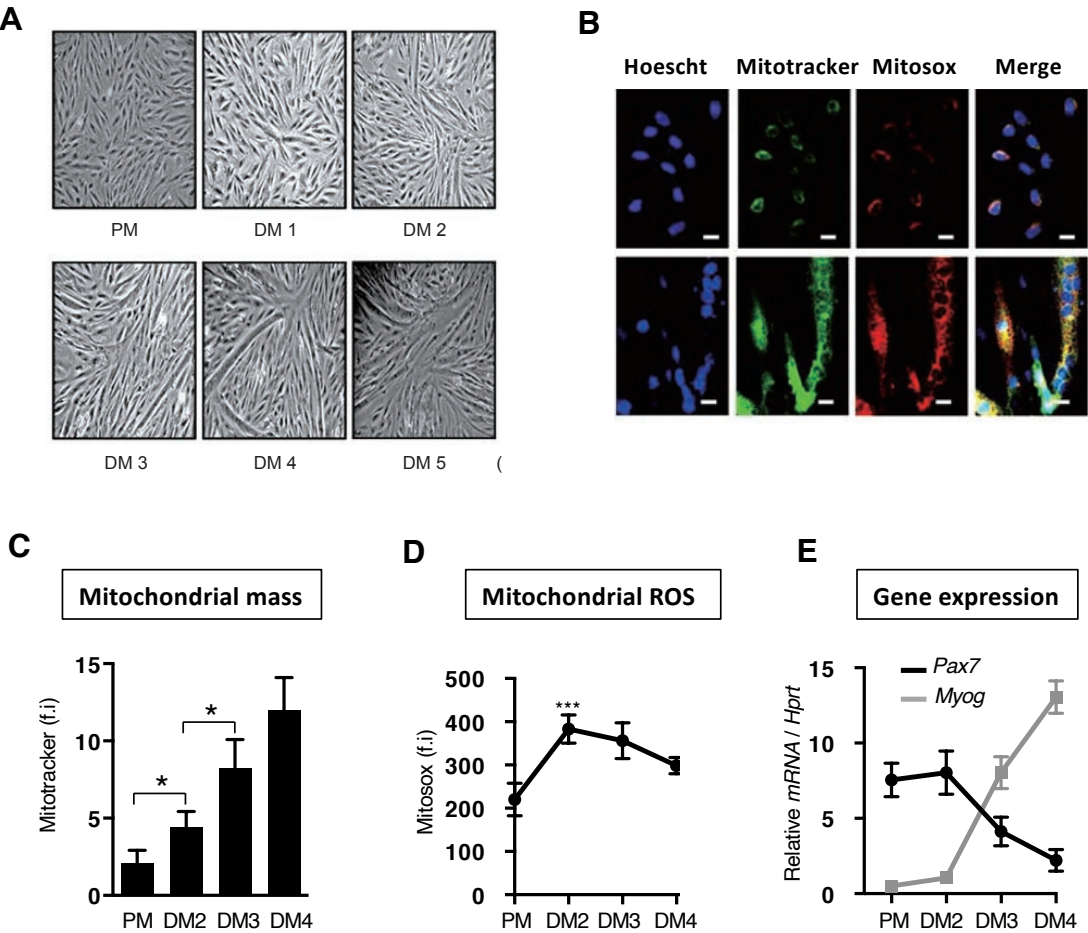
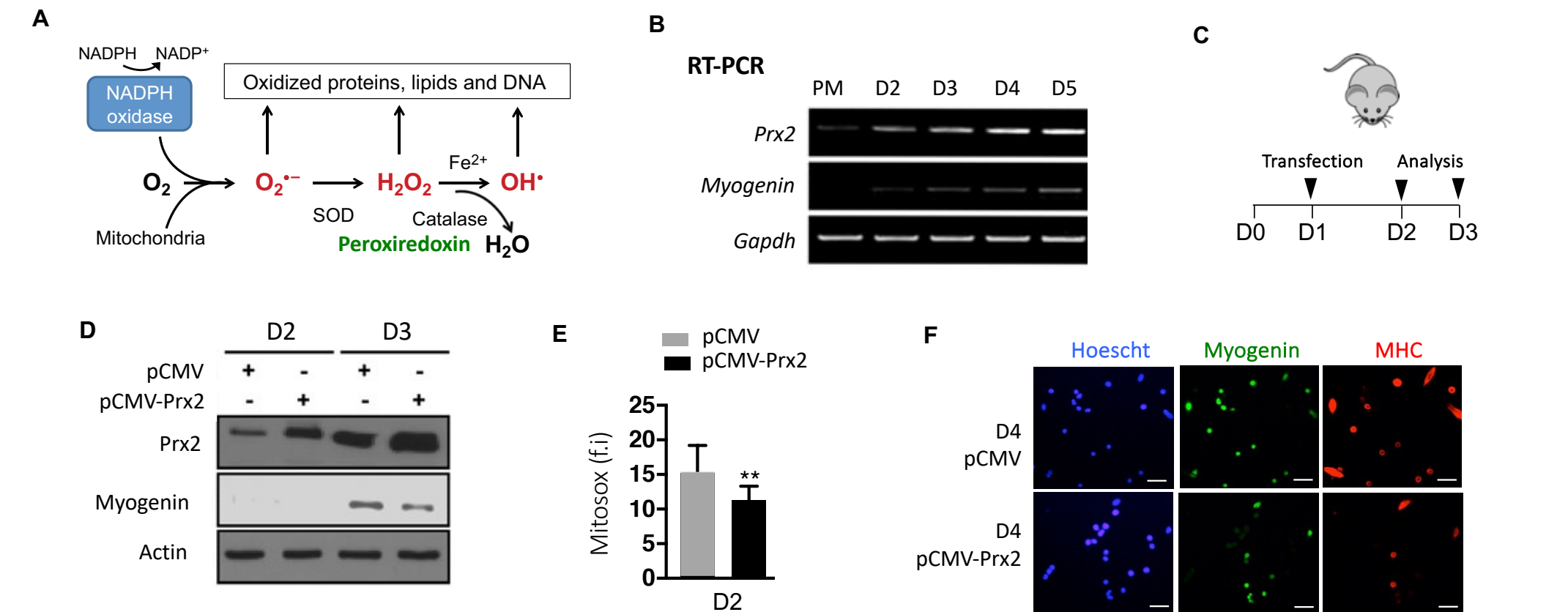


Figure 2a



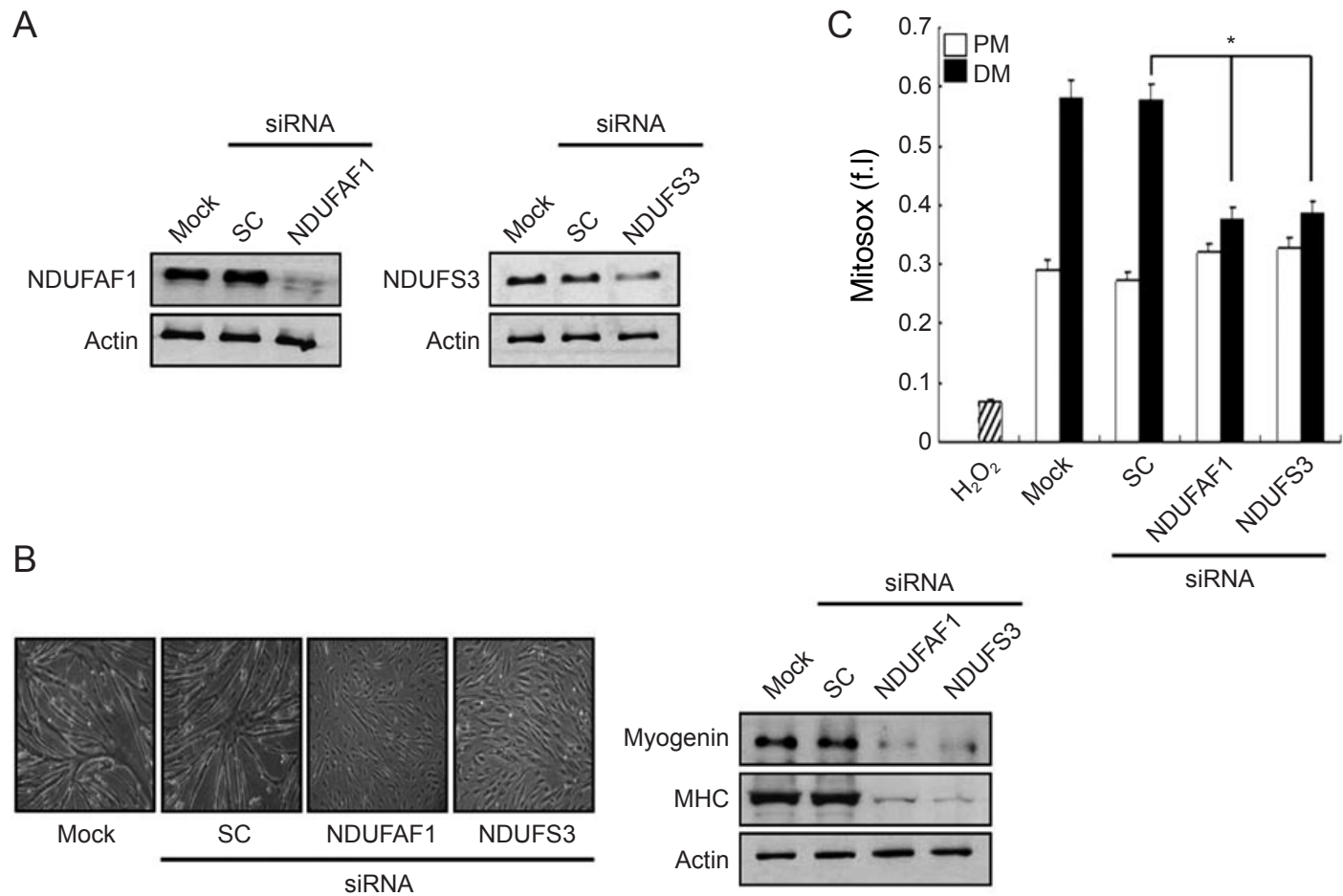
Muscle stem cells were purified from adult C57Bl6 mice and cultured for 6 days in proliferation (PM) and differentiation (DM) medium. **(A)** Morphological changes with progression of differentiation in DM for up to 5 days. **(B)** Mitochondrial mass and mROS generation were verified via confocal microscopy. Mitochondrial tracker DiOC6, green; mitochondrial ROS indicator MitoSOX, red; nuclear indicator Hoechst 33342, blue. Scale bar, 20 μ m. **(C, D)** Mitochondrial mass and mROS generation were quantified by flow cytometry using Mitochondrial tracker DiOC6 and mitochondrial ROS indicator MitoSOX in cultured cells at indicated days. **(E)** Cultured cells at indicated days were used for mRNA extraction. mRNAs were then assayed by qPCR using *Pax7* and *Myogenin* oligonucleotides.

Figure 2b



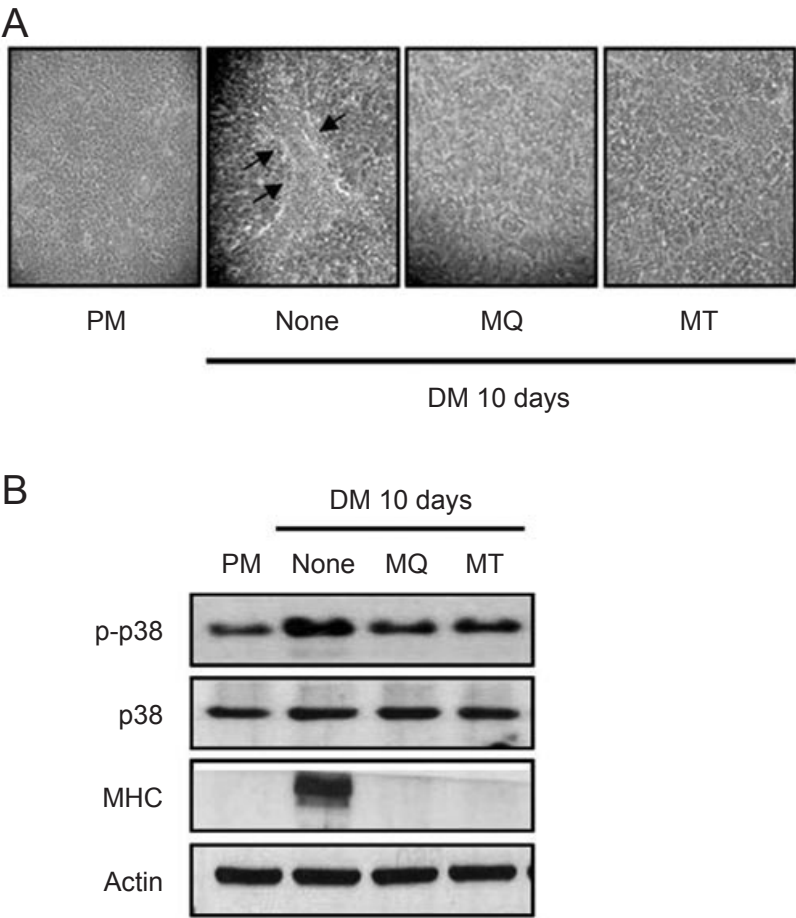
(A) Schematic representation of different Reactive Oxygen Species, and antioxidant enzymes. **(B)** Muscle stem cells purified from adult C57Bl6 mice were cultured in proliferation (PM) and differentiation (DM) medium, and used for mRNA extraction. mRNA were assayed by RT-PCR using *Prx2*, *Myogenin* and *Gapdh* oligonucleotides **(C, D, E, F)** Muscle stem cells purified from adult C57Bl6 mice were cultured and transfected at D1 with pCMV or pCMV-Prx2 constructs . **(D)** Cell lysates were subjected to SDS-PAGE followed by immunoblotting with anti-Prx2, anti-Myogenin and anti-Actin antibodies. **(E)** mROS generation were quantified by flow cytometry using mitochondrial ROS indicator MitoSOX. **(F)** Myogenin, Myosin heavy chain (MHC) expression were detected by immunofluorescence, using anti-Myogenin and anti-MHC antibodies. Nuclei were detected using Hoescht. Data represent means +/- SD, n=3.

Figure 3a



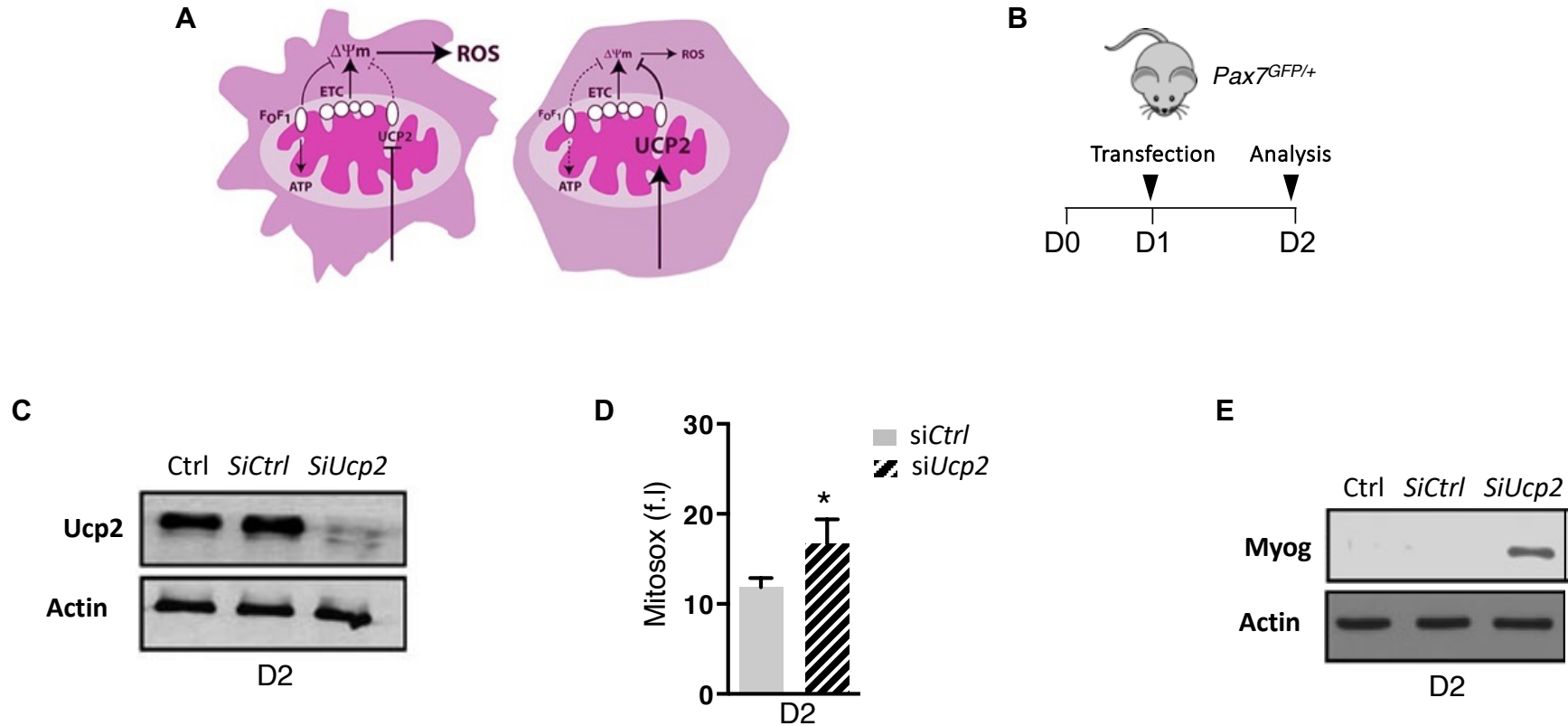
(A) Muscle stem cells purified from adult C57Bl6 mice were cultured and transfected with *SiCtrl*, *SiNDUFAF1* or *siNDUFS3*. Cells were then cultured in proliferation and differentiation medium and analyzed. **(B)** Morphological changes in DM for 5 days. Cell lysates were subjected to SDS-PAGE followed by immunoblotting with anti-Myogenin, anti-Myosin Heavy Chain (MHC) and anti-Actin antibodies. **(C)** mROS generation were quantified by flow cytometry using mitochondrial ROS indicator MitoSOX. Data represent means +/- SD, n=3.

Figure 3b



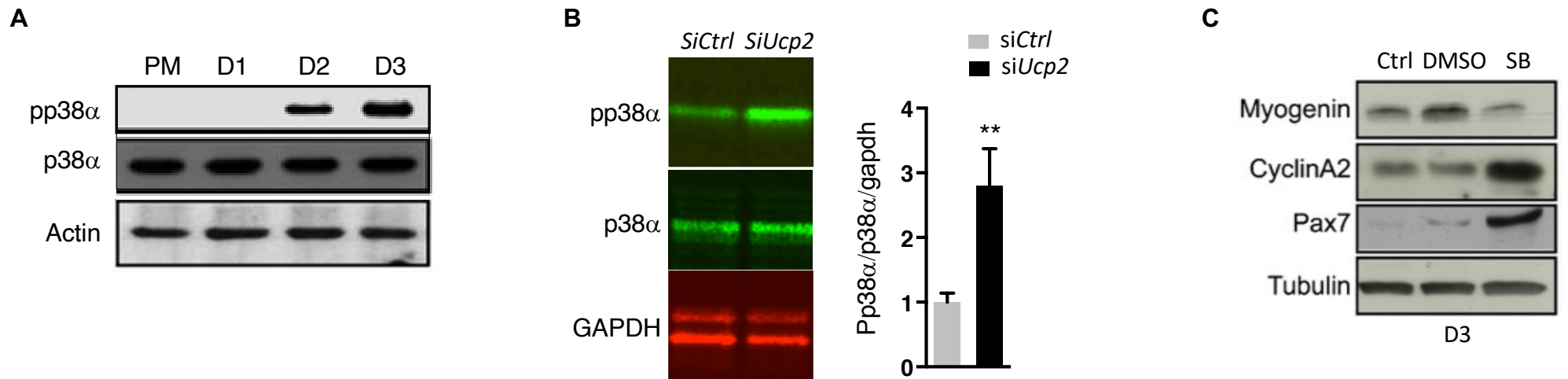
(A) Muscle stem cells purified from adult C57Bl6 mice were cultured in proliferation (PM) and differentiation medium (DM) and treated with 200 nM MitoQ (MQ) or 100 μ M MitoTempol (MT) in DM for 10 days. **(B)** Morphological changes observed in proliferation and differentiation medium in all conditions. **(B)** Cell lysates were subjected to SDS-PAGE followed by immunoblotting with anti-phospho-p38 α , anti-p38 α , anti-Myosin Heavy Chain (MHC) and anti-Actin antibodies.

Figure 4a



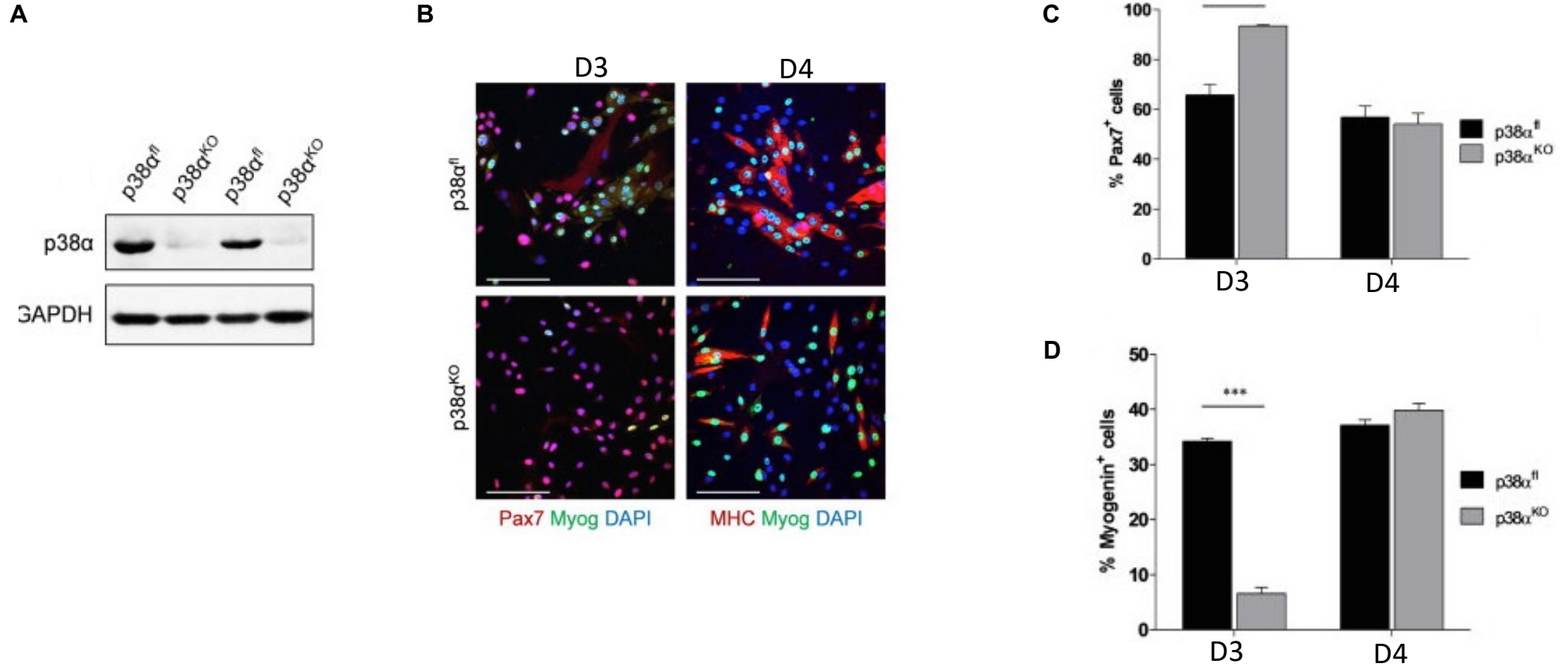
(A) Schematic representation of Ucp2 localization and function. **(B, C, D, E)** Muscle stem cells purified from adult C57Bl6 mice were cultured and transfected at D1 with *SiCtrl* or *SiUcp2*. Cells were then analyzed at days 2. **(C)** Cell lysates were subjected to SDS-PAGE followed by immunoblotting with anti-Ucp2 and anti-Actin antibodies. **(D)** mROS generation were quantified by flow cytometry using mitochondrial ROS indicator MitoSOX. **(E)** Cell lysates were subjected to SDS-PAGE followed by immunoblotting with anti-Ucp2 and anti-Actin antibodies. Data represent means $\pm$ SD, $n=3$.

Figure 4b



(A) Muscle stem cells purified from adult C57Bl6 mice were cultured in proliferation (PM) and differentiation (DM) medium. Cell lysates were subjected to SDS-PAGE followed by immunoblotting with anti-phospho-p38 α , anti-p38 α and anti-Actin antibodies. **(B)** Muscle stem cells transfected at D1 with *SiCtrl* or *SiUcp2* were used for protein extraction. Cell lysates were subjected to SDS-PAGE followed by immunoblotting with anti-phospho-p38 α , anti-p38 α and anti-GADH antibodies **(C)** Muscle stem cells purified from adult C57Bl6 mice were cultured in differentiation (DM) medium for 3 days and treated with DMSO or SB203580 (p38 chemical inhibitor). Cell lysates were subjected to SDS-PAGE followed by immunoblotting with anti-Myogenin, anti-Cyclin A2, anti-Pax7 and anti-Tubulin antibodies. Data represent means $\pm$ SD, n=3.

Figure 4c



(A) Muscle stem cells purified from adult control p38α^{flox/flox} or p38α^{-/-} mice were analyzed by SDS-PAGE followed by immunoblotting with anti-p38α and anti-GAPDH antibodies. **(B)** Muscle stem cells purified from adult control p38α^{flox/flox} or p38α^{-/-} mice were cultured for 3 and 4 days in differentiation medium and analyzed by immunofluorescence using antibodies directed against Pax7, Myogenin. Nuclei were stained with DAPI. **(C, D)** Cells positive for Pax7 (C) and Myogenin (D) expression were counted. Their number was represented as a percentage of total population. Data represent means $\pm$ SD, n=3.